

Classification with Linear Models

Losses for linear classification, logistic regression, multiclass classification

Data Analytics and Mining, 2025

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November 21, 2025

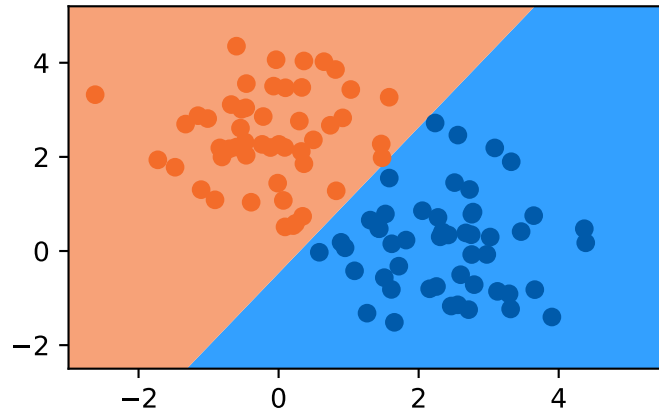
Can't we just use linear regression
for classification?



Classification with linear regression

Classification:

$$\hat{f}(x) = \text{sign}[\theta^T x]$$



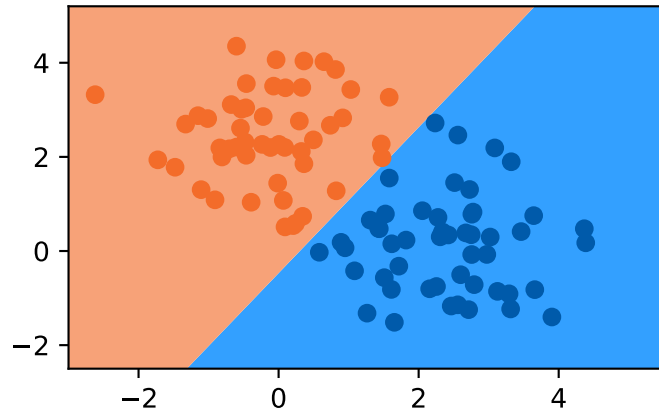
For binary classification task, assign:

- $y = +1$ for **positive** class
- $y = -1$ for **negative** class

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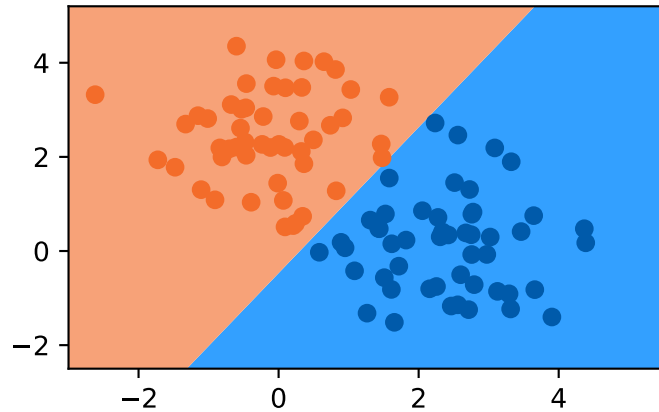
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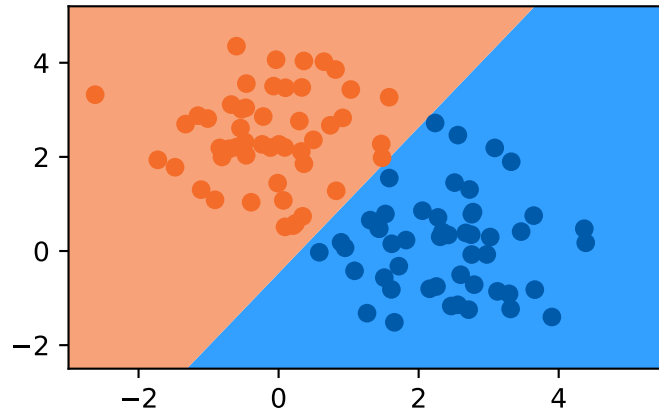
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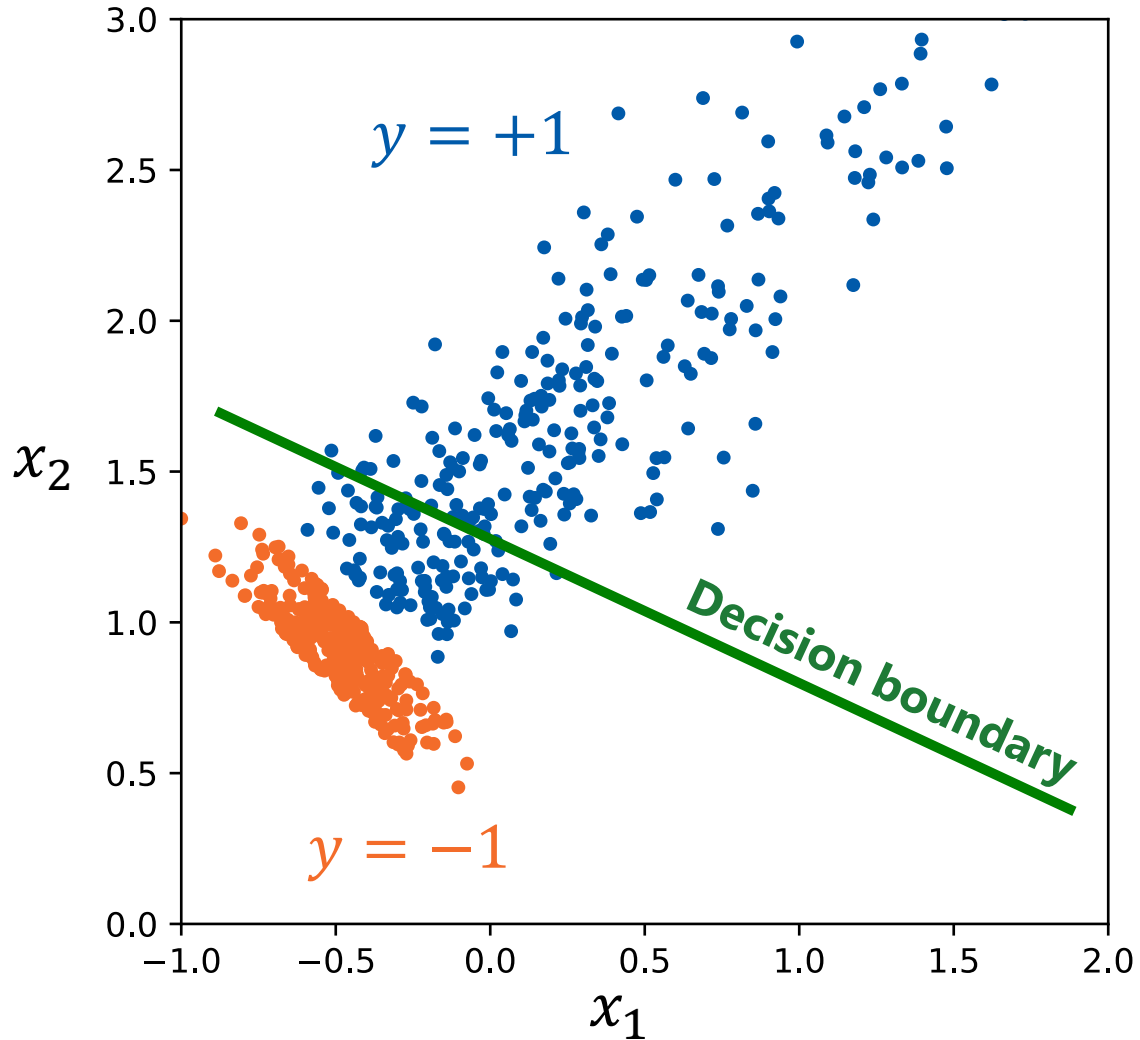
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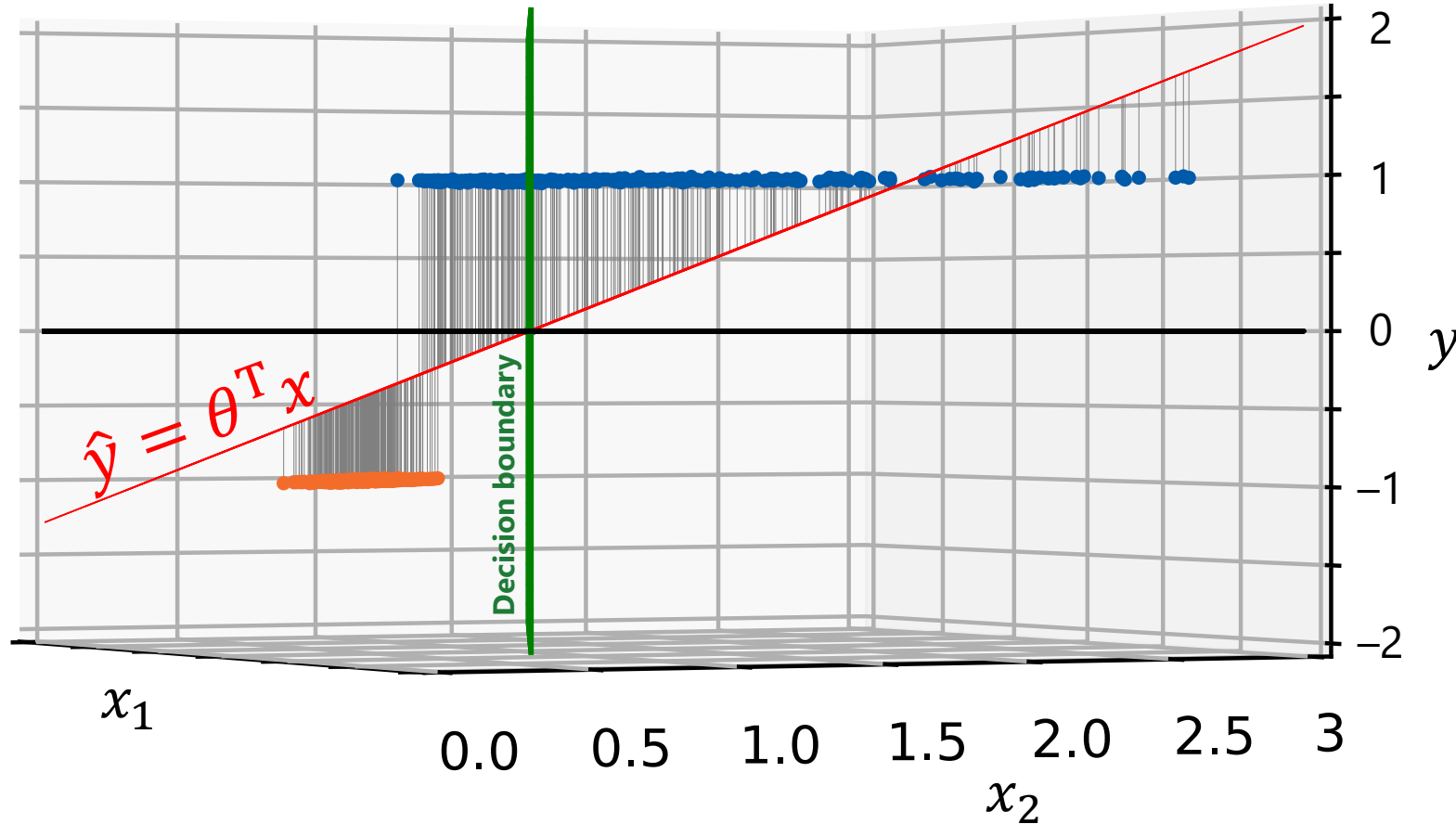
Any problems with this approach?

Classification with linear regression



May face problems when classes are unbalanced or have different spread

Classification with linear regression



MSE loss makes the model **avoid high residuals**

at a price of **pushing the decision boundary** towards the class with higher spread

Can we find a better loss function?

Classification loss functions

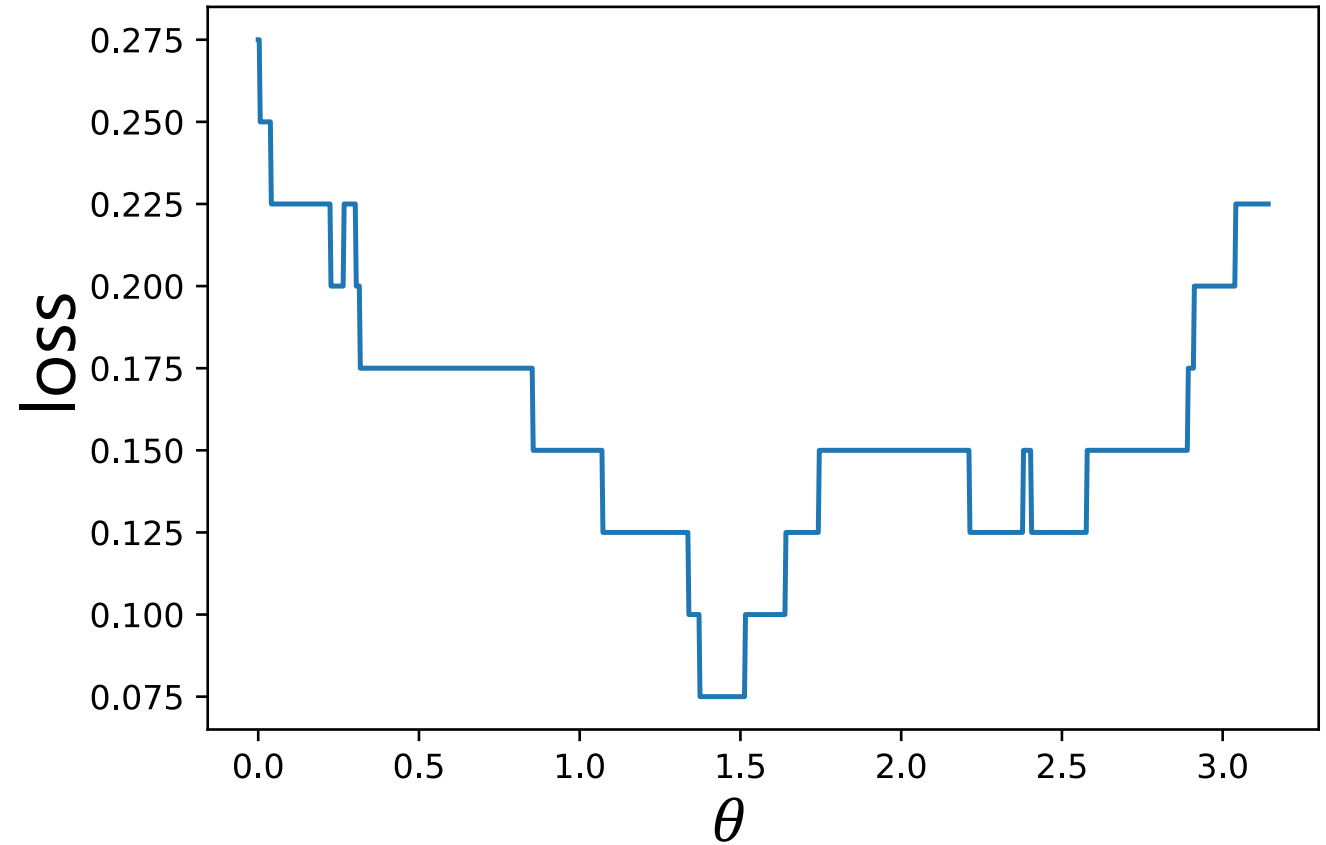


0-1 Loss

Probability of an error

$$\mathcal{L}_{0-1} = \frac{1}{N} \sum_{i=1 \dots N} \mathbb{I}(\theta^T x_i \cdot y_i < 0)$$

$$y_i \in \{-1, +1\}$$

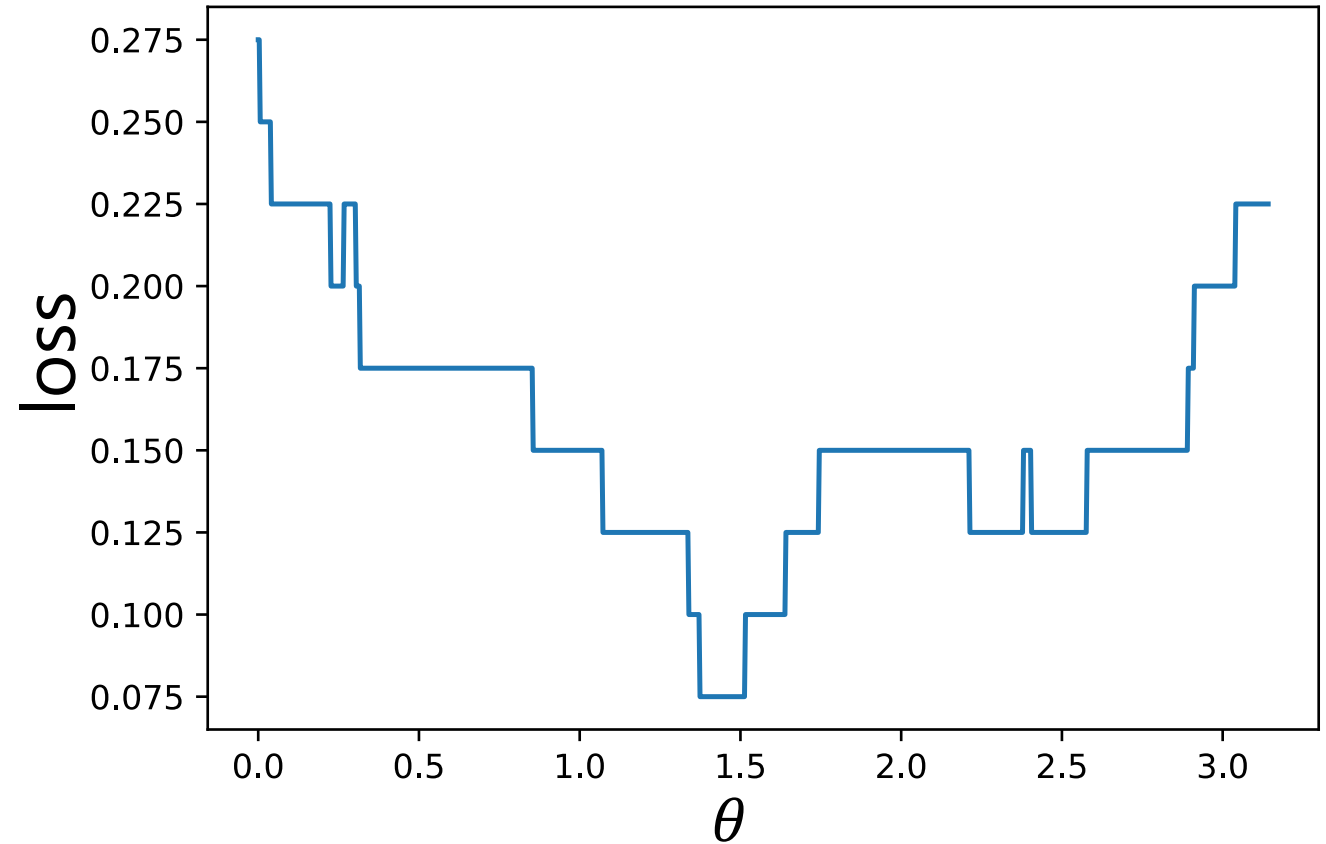


0-1 Loss

Probability of an error

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Can't optimize **piecewise constant** function with gradient-based methods*

*other techniques exist (still quite limited)

Margin

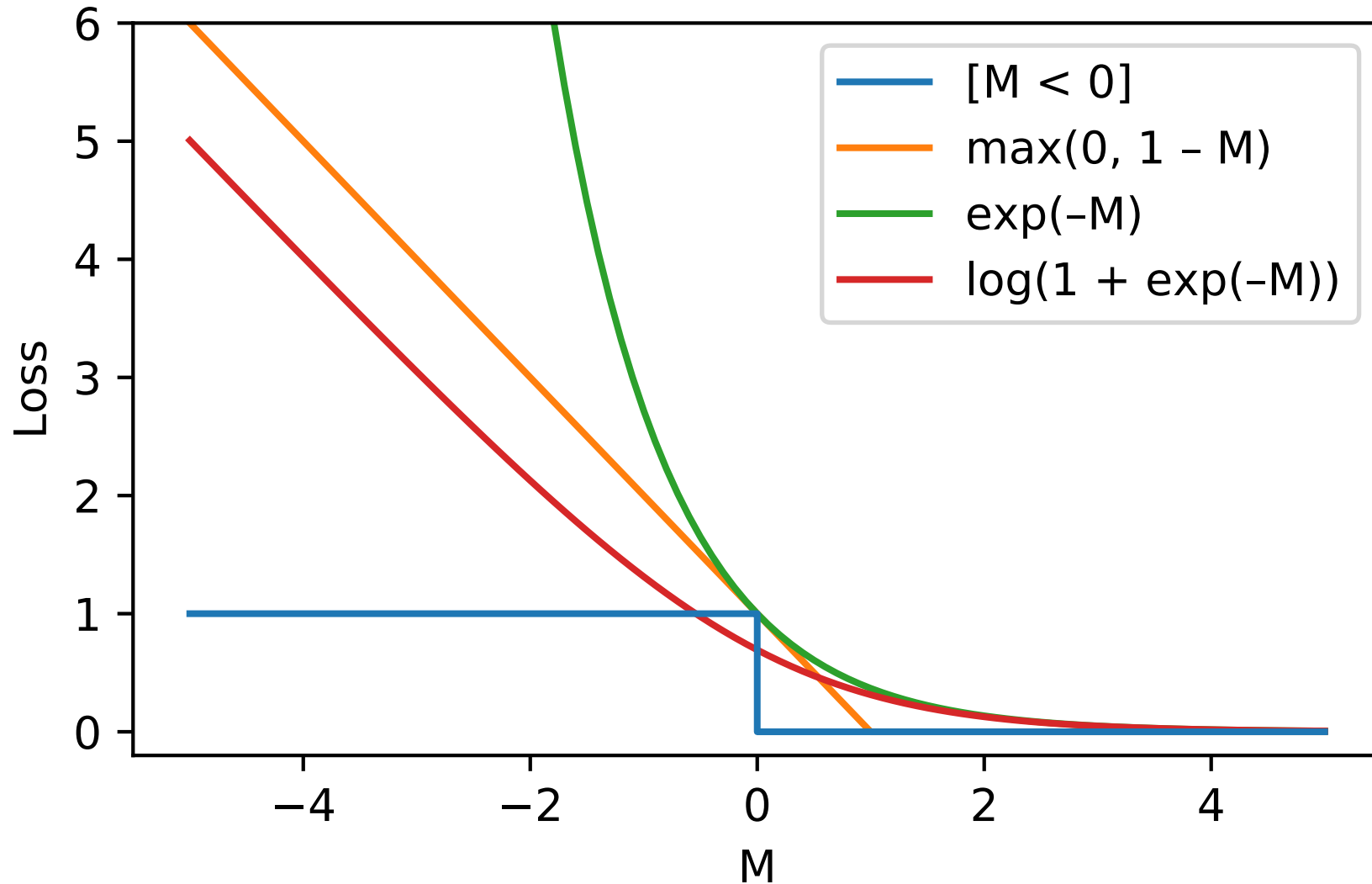
$$M = \theta^T x \cdot y$$

$$\mathcal{L}_{0-1} = \frac{1}{N} \sum_{i=1 \dots N} \mathbb{I}(\underbrace{\theta^T x_i \cdot y_i}_{\text{margin}} < 0)$$

$M > 0$ – correct classification

$M < 0$ – incorrect classification

Upper bounds on 0-1 loss



Instead of optimizing the 0-1 loss we can optimize a **differentiable upper bound**

Logistic Regression



Linear probability model

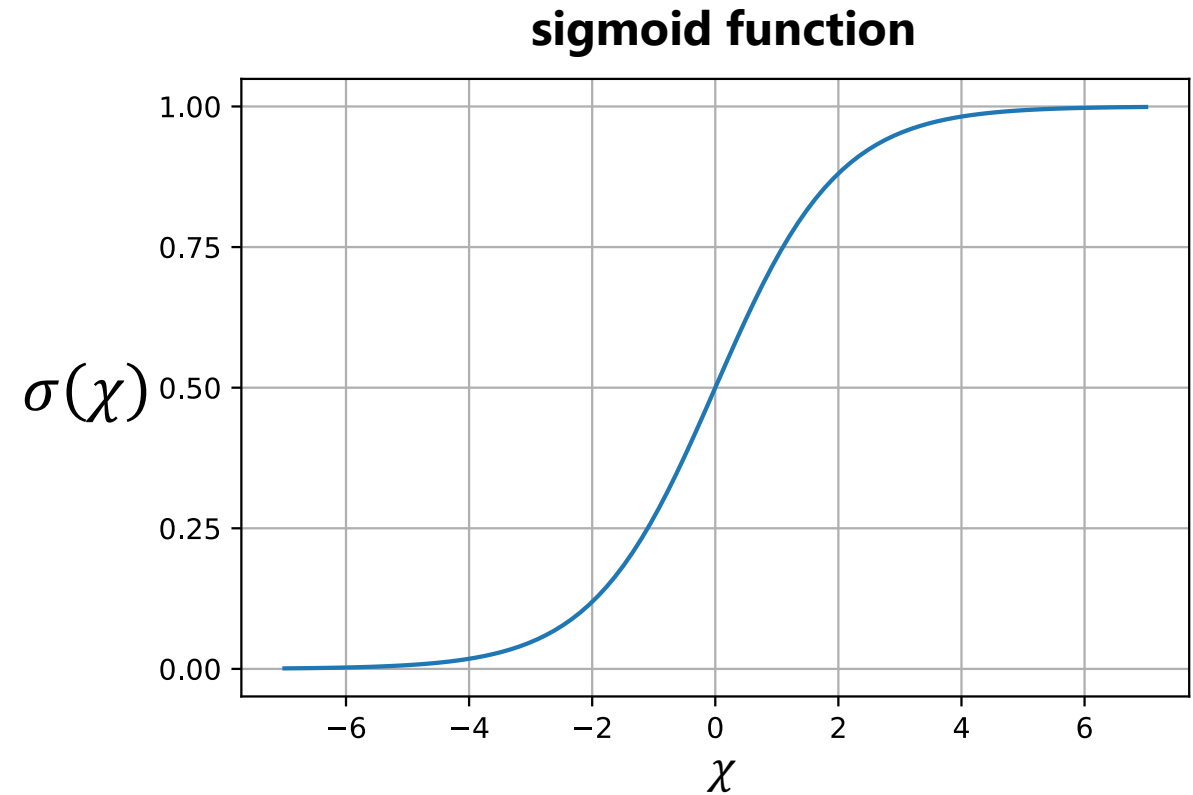
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Linear probability model

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Common choice – **sigmoid function**:

$$\sigma(\chi) = \frac{1}{1 + e^{-\chi}}$$



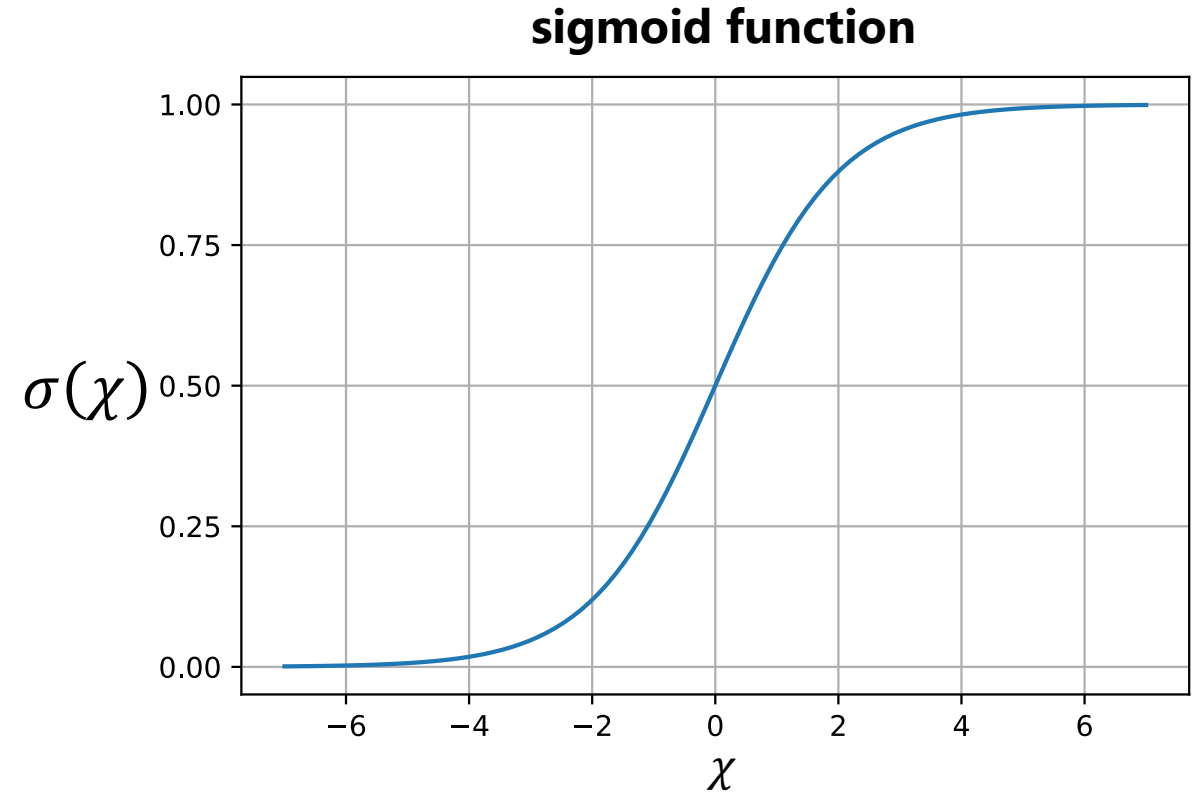
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i.e. $P(y = +1|x) = \sigma(\theta^T x)$



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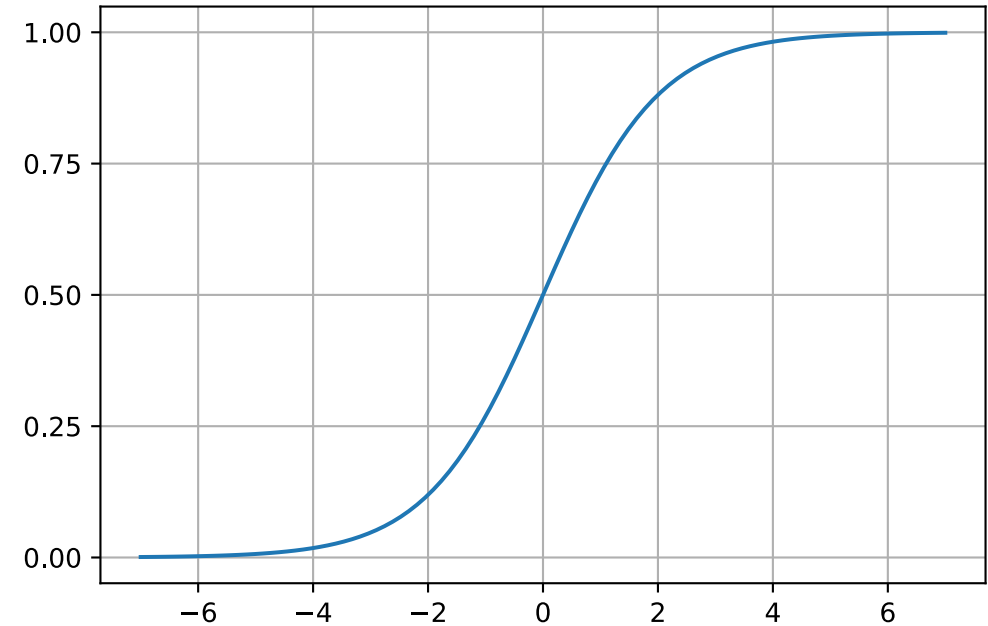
$\sigma(\chi)$

Use negative log likelihood as our loss function:

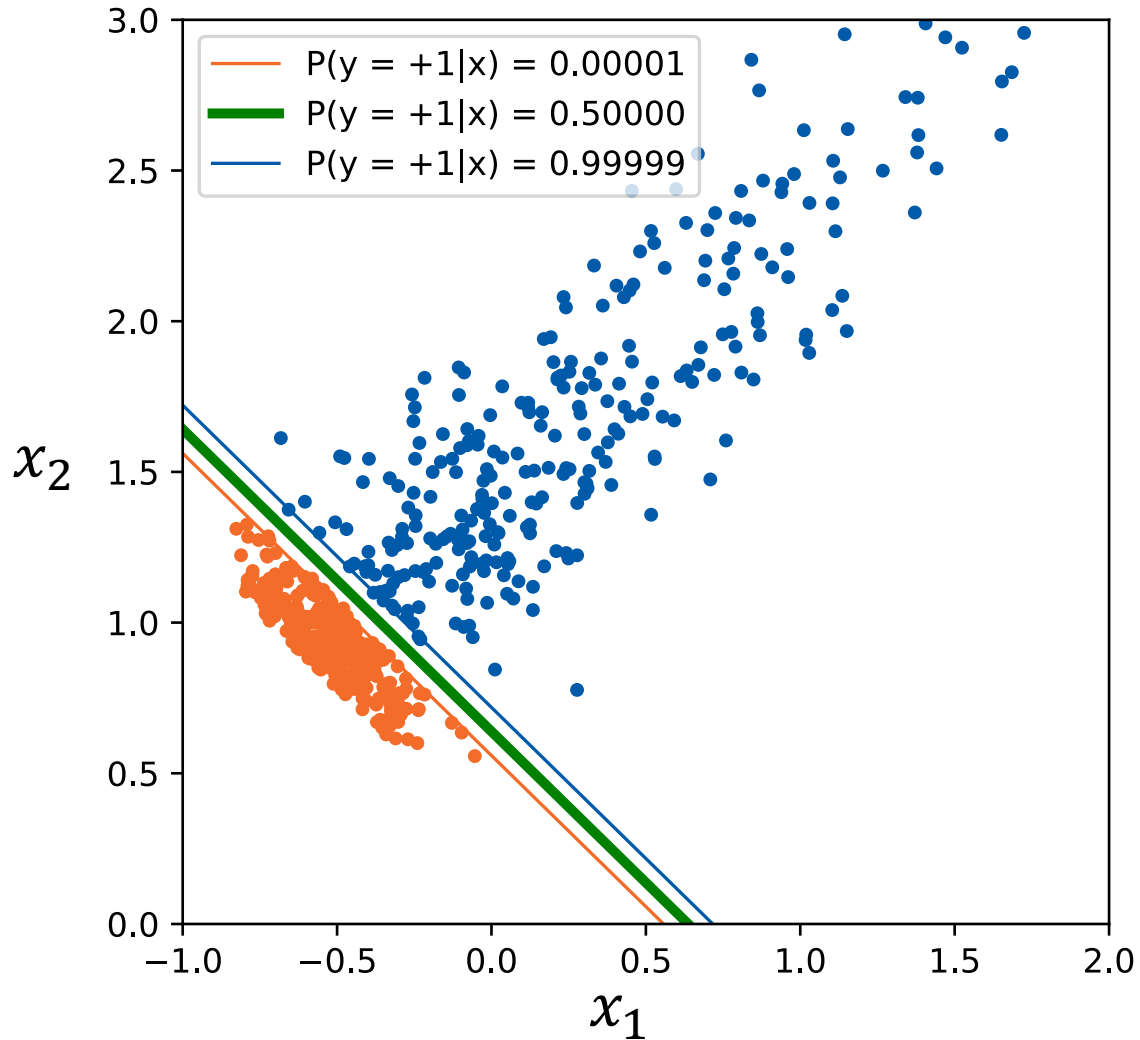
$$\mathcal{L} = - \sum_{i=1 \dots N} \log \sigma(\theta^T x_i \cdot y_i) = \sum_{i=1 \dots N} \log (1 + e^{-\theta^T x_i \cdot y_i})$$

This can be optimized **numerically**

sigmoid function

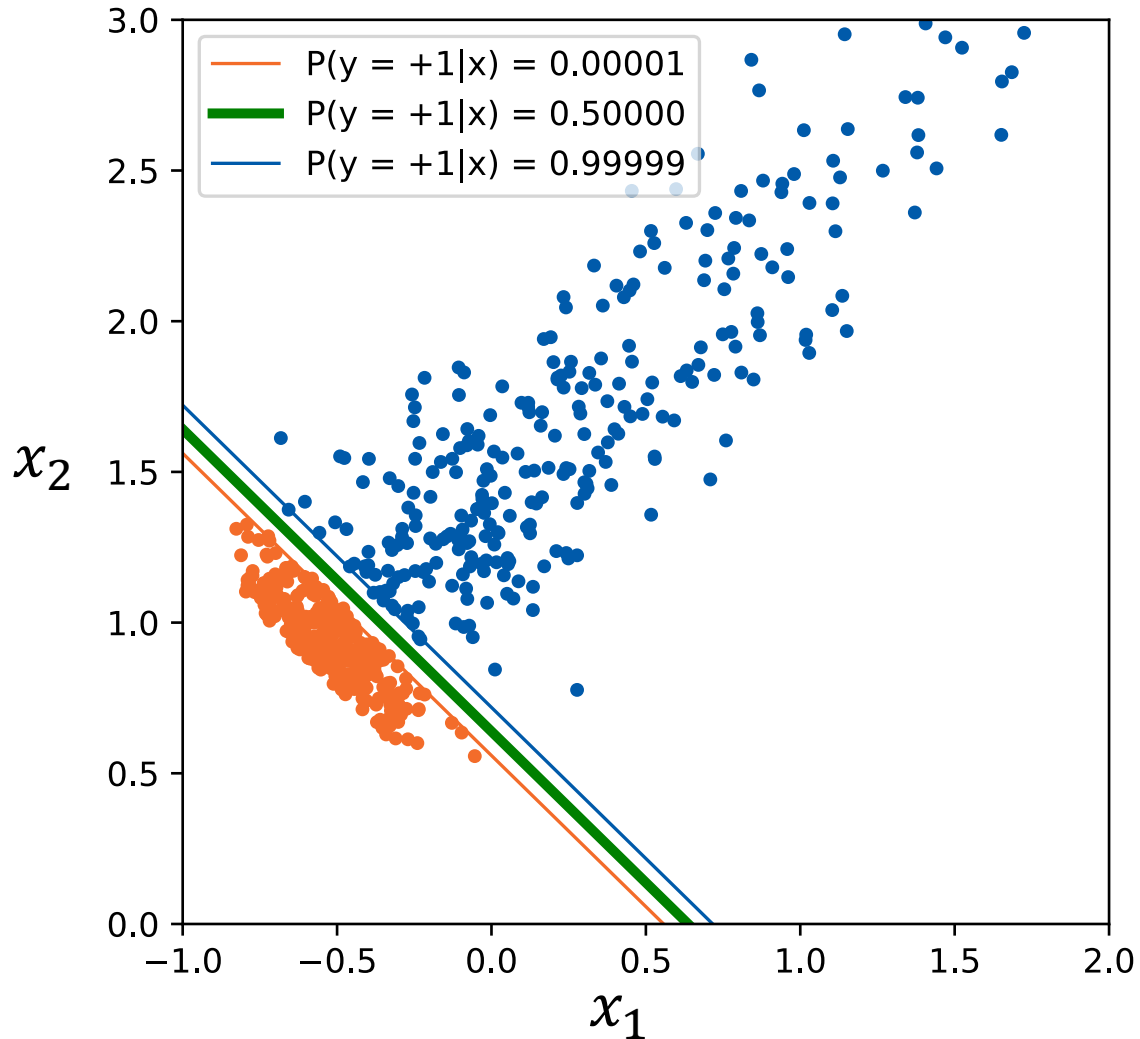


Example



Now the boundary is at the right place

Example



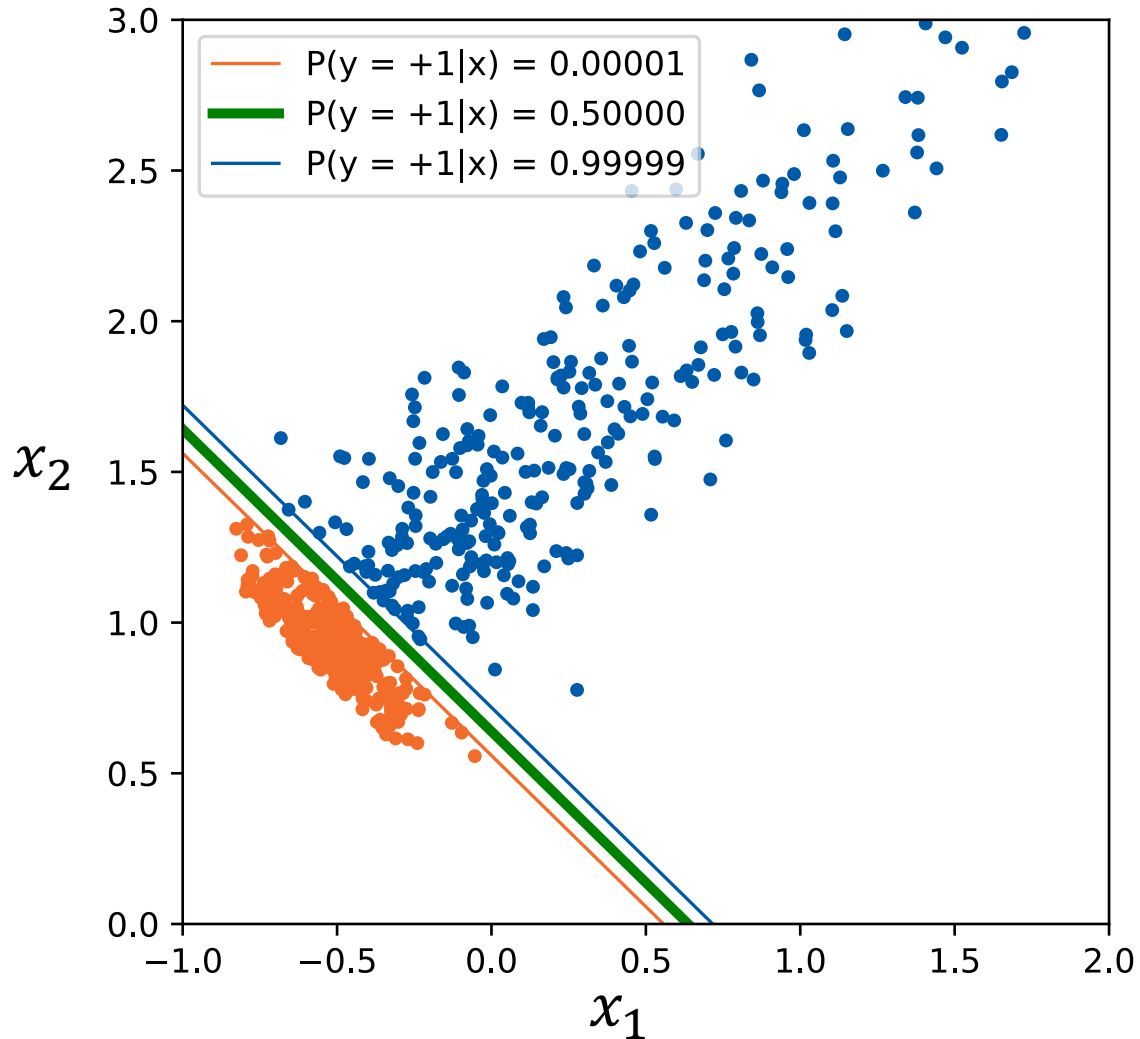
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Note: when classes are linearly separable for any correct decision boundary

$$\theta \rightarrow C \cdot \theta, \text{ for some } C > 1 \in \mathbb{R}$$

keeps the boundary at the same place, yet improves the loss

Example



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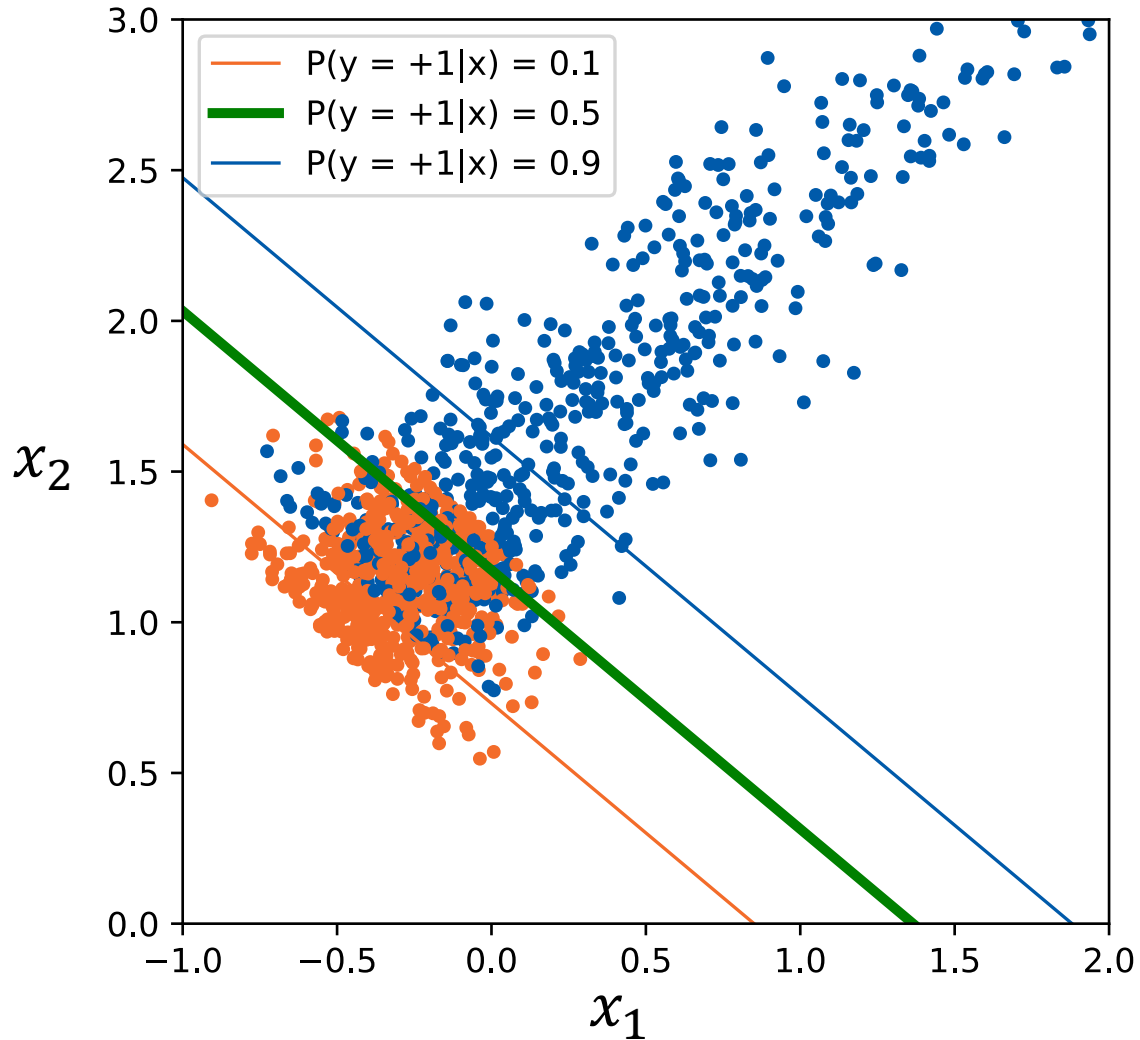
Note: when classes are linearly separable for any correct decision boundary

$$\theta \rightarrow C \cdot \theta, \text{ for some } C > 1 \in \mathbb{R}$$

keeps the boundary at the same place, yet improves the loss

ideal fit when sigmoid turns into a step function (at infinitely large θ)

Example



When classes overlap the loss has a finite minimum

Predicted class probability changes smoothly

Multiclass Logistic Regression



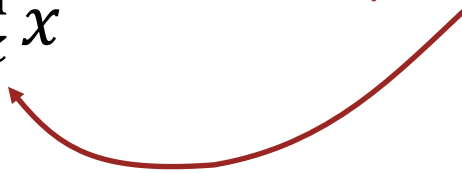
Multinomial Logistic Regression

Similarly to the binary case, we'll model the class probabilities

Let's model **unnormalized** class probabilities like this:

$$\tilde{P}(y = k|x) = \exp \theta_k^T x$$

Note: now we have K
parameter vectors



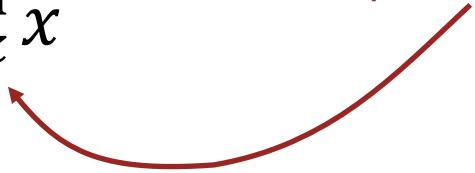
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Then, the **normalized** probabilities are:

$$P(y = k|x) = \frac{\tilde{P}(y = k|x)}{\sum_{k'=1 \dots K} \tilde{P}(y = k'|x)} = \frac{\exp \theta_k^T x}{\sum_{k'=1 \dots K} \exp \theta_{k'}^T x}$$

- This function is called **softmax** and is commonly used in neural networks

Multinomial Logistic Regression

Plugging everything into the negative log likelihood we get our loss function:

$$\mathcal{L} = - \sum_{i=1 \dots N} \log \frac{\exp \theta_{y_i}^T x_i}{1 + \sum_{k'=1 \dots K-1} \exp \theta_{k'}^T x_i}$$
$$(\theta_K = 0)$$

Again, this can be optimized **numerically**

Multiclass classification: general approach



General idea

For a problem with K classes introduce K predictors:

$$\hat{f}_k(x): \mathcal{X} \rightarrow \mathbb{R}, \text{ for } k = 1, \dots, K$$

each of which outputs a corresponding **class score**.

Predict the class with the **highest score**:

$$\hat{y}_i = \operatorname{argmax}_k \hat{f}_k(x_i)$$

Example: binary → multiclass

Any binary linear classification model can be converted to multiclass with **one-vs-rest** strategy

Example: binary $\rightarrow$ multiclass

Any binary linear classification model can be converted to multiclass with **one-vs-rest** strategy

For each class k train a binary model $\hat{f}_k(x) = \theta_{(k)}^T x$ separating the given class from all others, $\hat{y}_{(k)}^{1\text{-vs-rest}} = \text{sign}[\hat{f}_k(x)]$

Example: binary $\rightarrow$ multiclass

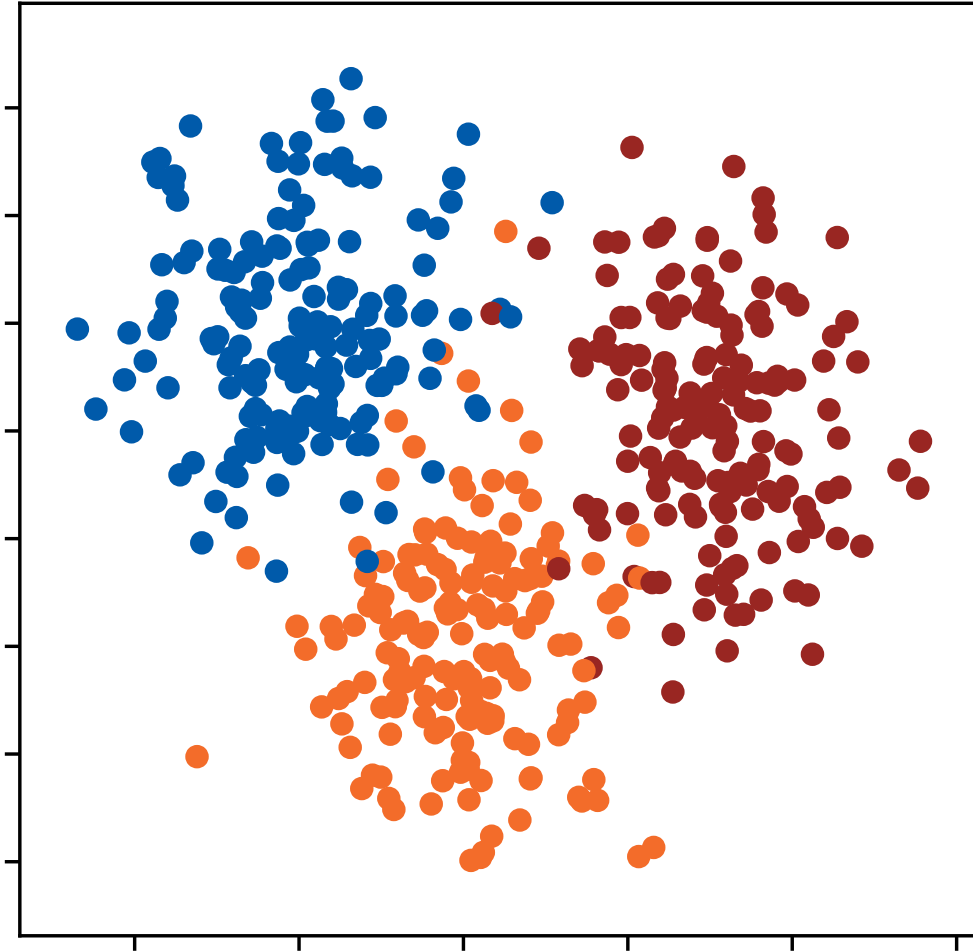
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Use the outputs of $\hat{f}_k$ as class scores for multiclass classification:

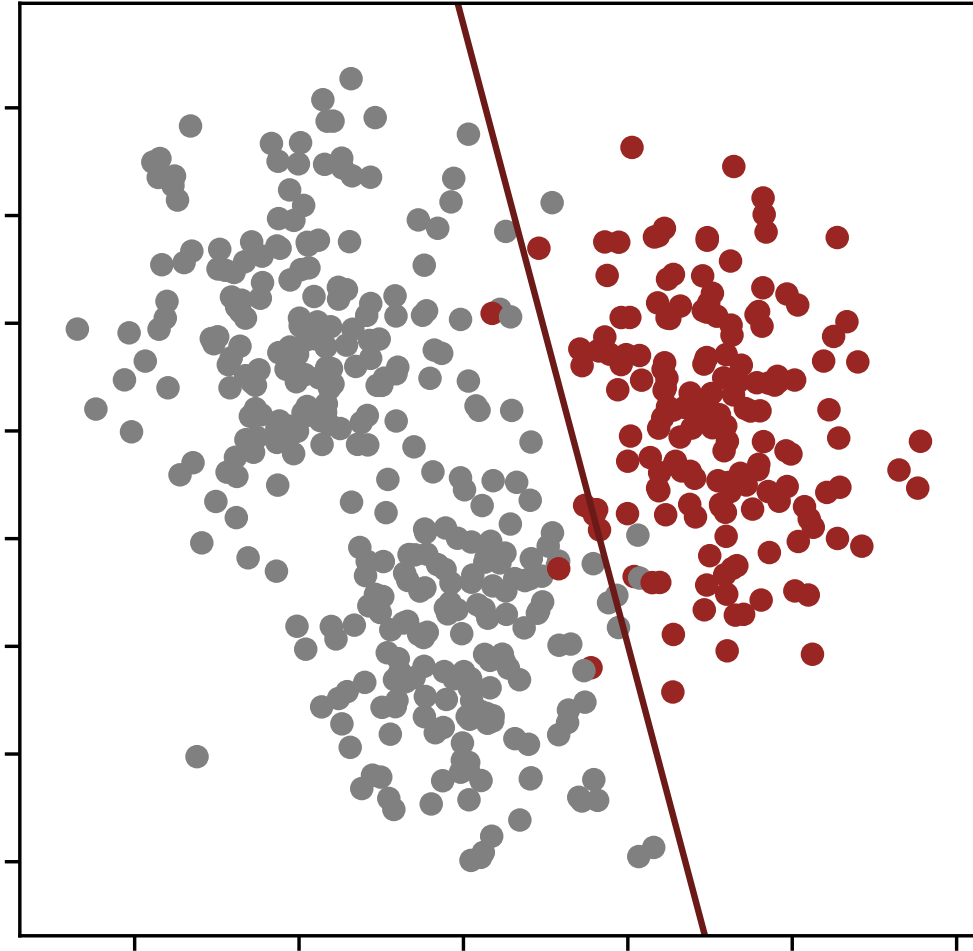
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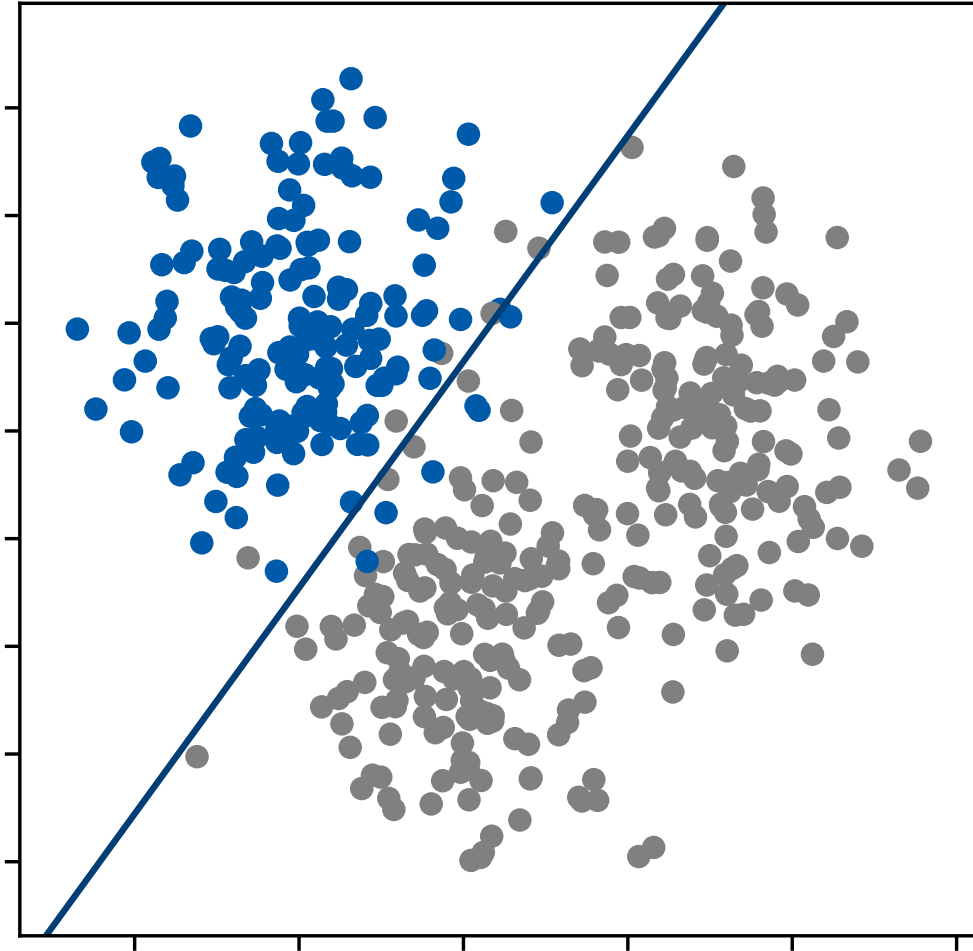
Consider the following 3 class problem

Example



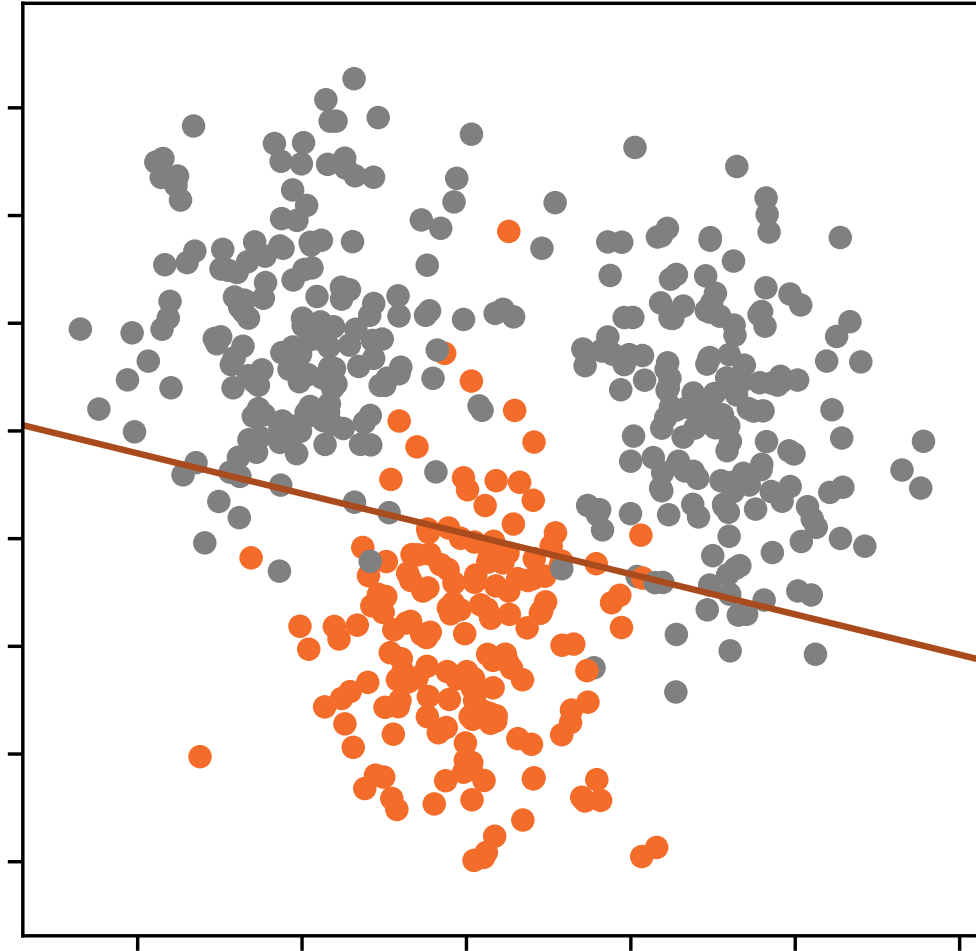
“Class-1 VS rest” binary classifier

Example



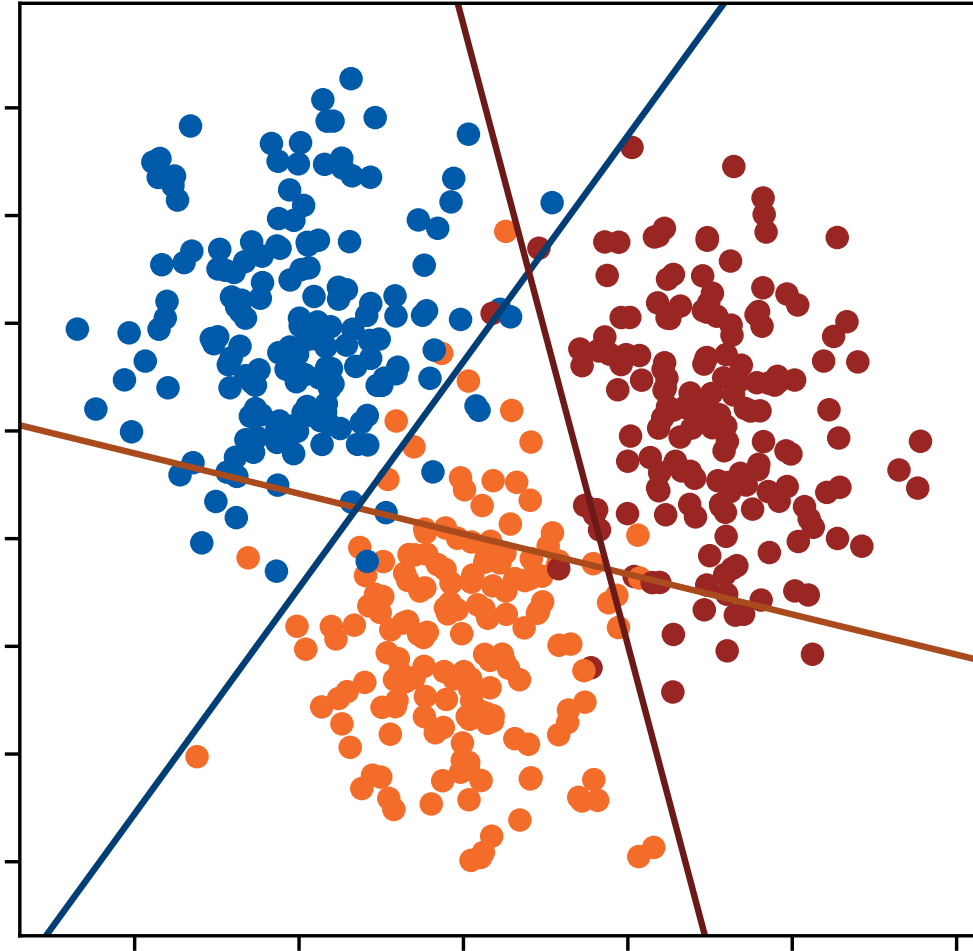
“Class-2 VS rest” binary classifier

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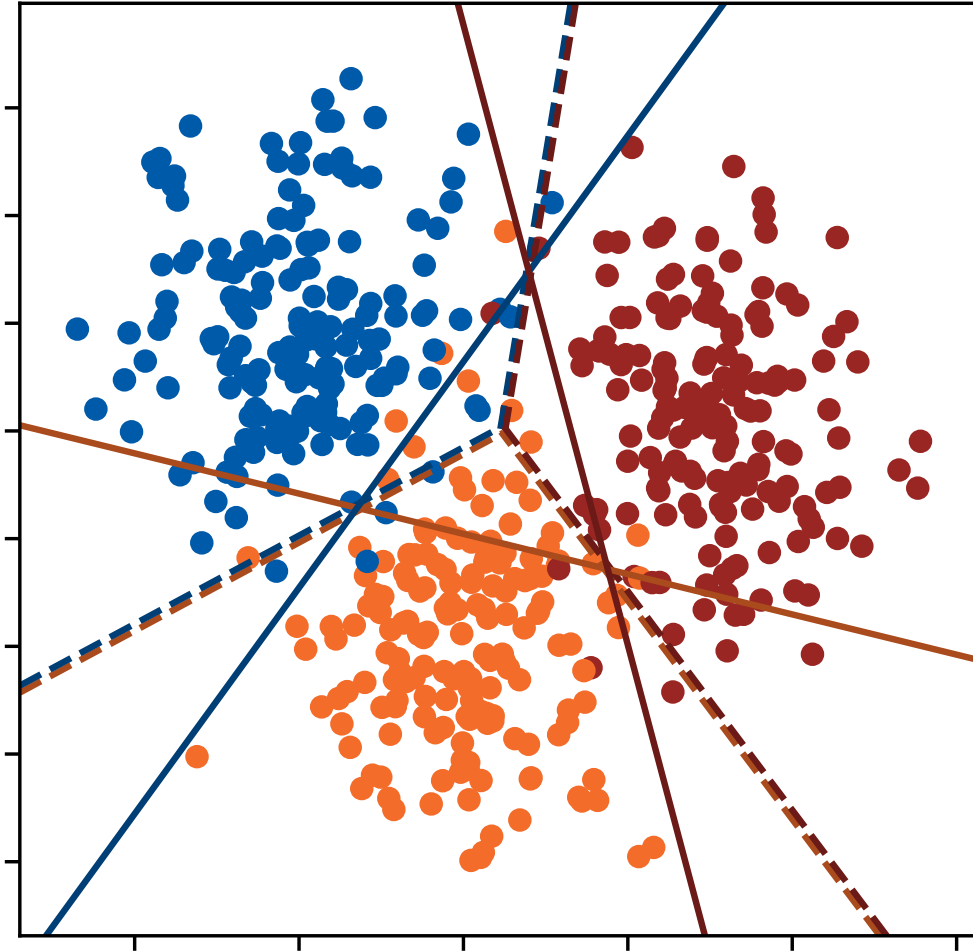
"Class-3 VS rest" binary classifier

Example



$\hat{f}_k(x) = 0$ lines (binary decision boundaries)

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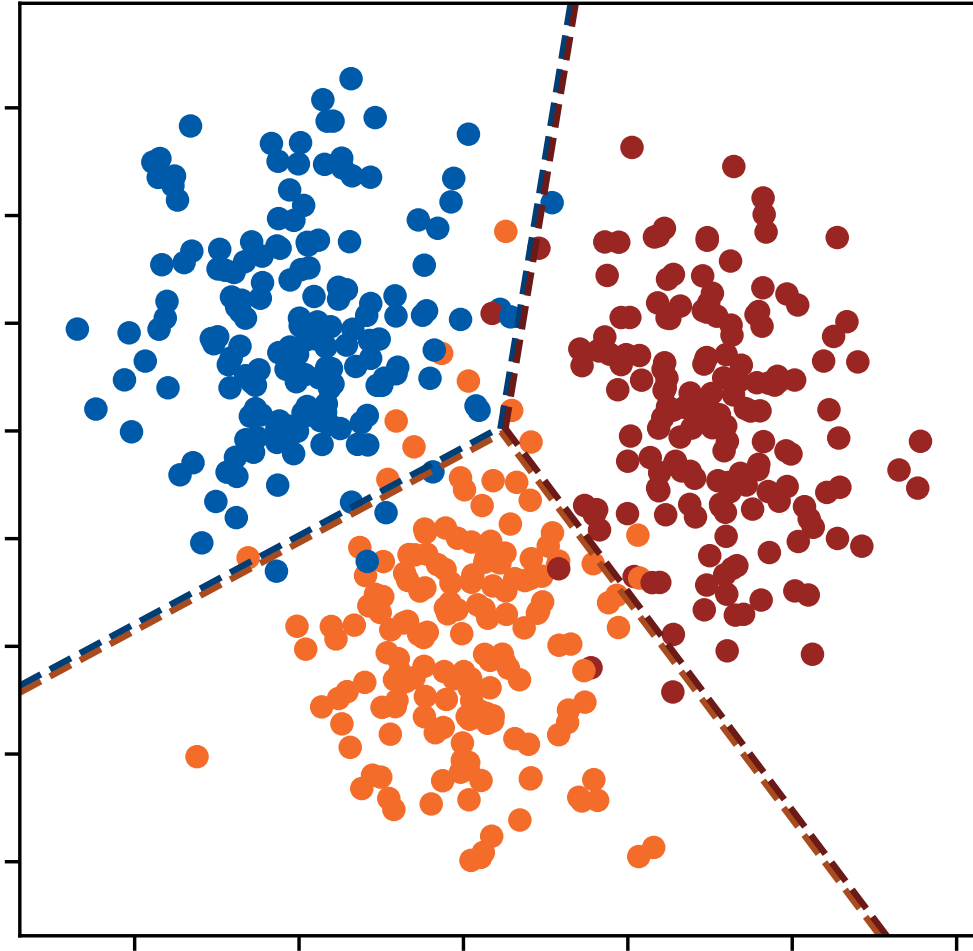


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Adding decision boundaries for

$$\hat{y} = \operatorname{argmax}_k \hat{f}_k(x)$$

Example



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Summary

Classification with linear regression and MSE loss may provide **biased results**

0-1 loss function is better, but is **hard to optimize** directly

Various **differentiable upper bounds** on 0-1 loss may be used instead

Logistic Regression combines such an upper bound with a **probabilistic model** using the **sigmoid function**

Generalizing sigmoid function to a multiclass case yields **softmax function**

Any binary linear classifier can be adapted to multiclass with the **one-vs-rest strategy**

Food for thought: how can you mitigate the biased probability problems when using one-vs-rest strategy (as discussed on the previous slide)?

Thank you!

Majid Sohrabi



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